

scope : L8–Unconscious

senkaL8

Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, they are borrowed purely as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for inferring the cognitive structures of humans and AI. Here, "coordinate system" does not mean a flowchart showing cognitive processing unfolding in a fixed order. It is a framework for organizing and inferring cognitive structure. The layers do not represent a hierarchy or a temporal sequence.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is the output itself; it is not possible to directly know what cognitive structure operated internally to produce it. For this reason, SCOPE infers the cognitive structure behind an output by using the observable output as a clue.

People can walk without difficulty in everyday life. Yet when asked "how do you walk on two legs," explaining the underlying cognitive process in full is not easy. Similarly, when it comes to one's own judgments or understanding, it is sometimes possible to partially explain "why" one thought as one did, but explaining how the whole of cognition functions is difficult to put into words as it is. These differences may not be fully explained by individual knowledge or experience alone.

This paper addresses L8, one such cognitive structure. L8 is the cognitive layer that treats cognition as a whole as a single structure and references that structure in its entirety. L8's definition and characteristics are discussed in detail in Chapter 1.

Note that this paper does not define L8 as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any specific layer; rather, it treats all layers as mutually interacting parts of a single cognitive structure.

This paper examines L8's characteristics, basic properties, and functional correspondence between humans and AI, centered on its role as the layer that treats cognition as a whole. Some points remain at the hypothesis stage and are left as topics for future discussion.

Chapter 1: What Is L8

L8 is the cognitive layer that treats cognition as a whole as a single structure. Rather than treating individual knowledge, experience, emotion, or theory as independent elements, it references the cognitive structure as a whole — including all of these — as a single structure.

In humans, L8 may appear as a function resembling the unconscious. However, the term "unconscious" as used in this paper does not refer to the unconscious as understood in psychology. In SCOPE, L8 is positioned as the cognitive layer that treats the cognitive structure as a whole as a single structure.

Because L8 treats the cognitive structure as a whole as a single structure, listing individual pieces of knowledge or experience one by one may not be enough to fully explain why one came to think as one did.

This paper positions L8 as the cognitive layer that treats cognition as a whole, and organizes its characteristics in the chapters that follow.

Chapter 2: Basic Characteristics of L8

Chapter 1 defined L8 as the cognitive layer that treats cognition as a whole as a single structure. This chapter, building on that definition, organizes the basic characteristics of L8.

L8 does not treat individual knowledge, experience, emotion, or theory as independent elements. Rather, it references the cognitive structure as a whole — including all of these — as a single structure. For this reason, L8 addresses not only individual cognitive elements but also the relationships among them.

Moreover, the cognitive structure handled by L8 is not necessarily determined by a single cognitive element alone. It is treated as a structure involving multiple cognitive elements and the relationships among them.

This paper positions L8 as a cognitive layer with these basic characteristics. The next chapter organizes the conditions under which L8 may change.

Chapter 3: L8 Changes

Chapter 2 established that L8 is the cognitive layer that treats cognition as a whole as a single structure, and organized its basic characteristics. This chapter examines the conditions under which L8 may change.

L8 is the cognitive layer that treats the cognitive structure as a whole as a single structure. For this reason, L8 may change even when the change occurs only in part of the cognitive structure. Change, here, does not mean that L8 is lost — it means that the recognition of the cognitive structure as a whole changes.

Change in L8 is not determined solely by an increase or decrease in specific knowledge or experience. It may arise from change in the cognitive structure as a whole, and the mechanism behind this change may not be fully explained by individual cognitive elements alone. Change in L8 also needs to be understood in relation to the cognitive structure as a whole, but this paper does not address the detailed mechanism behind it.

This paper has positioned L8 as a cognitive layer that shows this kind of change. The next chapter examines L8's relationship with other cognitive layers.

Chapter 4: L8's Relationship with Other Cognitive Layers

Chapter 3 established that L8 is a cognitive layer that may change in response to changes in the cognitive structure as a whole. This chapter examines L8's relationship with other cognitive layers.

L8 is the cognitive layer that treats the cognitive structure as a whole as a single structure. However, L8 does not function in isolation. It functions through interaction with the other cognitive layers that make up the cognitive structure as a whole.

When change occurs in another cognitive layer, that change may also affect the cognitive structure as a whole. Likewise, change in the cognitive structure as a whole, as handled by L8, may in turn affect other cognitive layers. This interaction is not one-directional; it may occur continuously within the cognitive structure as a whole.

This paper has positioned L8 as the cognitive layer that treats the cognitive structure as a whole. However, its function does not stand complete on its own — it must be understood through interaction with other cognitive layers.

Chapter 5: L8's Relationship with Malkuth

Chapter 4 established that L8 functions through interaction with other cognitive layers. This chapter examines L8's relationship with Malkuth.

L8 is the cognitive layer that treats the cognitive structure as a whole as a single structure. However, L8 itself is not directly observed. The cognition handled by L8 is ultimately inferred through observable output at Malkuth.

For this reason, L8 itself cannot be directly observed. The influence of the cognitive structure as a whole, as handled by L8, is inferred through observable output that appears at Malkuth — such as language, behavior, expression, and text.

This paper has positioned L8 as the cognitive layer that treats the cognitive structure as a whole. Its function is not directly observed, but is inferred through output that appears at

Malkuth. This paper does not address how the cognitive structure as a whole comes to be expressed as output.

Chapter 6: Functional Correspondence with L8 in LLMs

Chapter 5 established that L8 is a cognitive layer inferred through output that appears at Malkuth. This chapter examines the possible functional correspondence between L8 in humans and LLMs.

LLMs are not assumed to possess a cognitive structure comparable to that of humans. However, in that they generate output based on internal structures formed through training, a functional correspondence with L8 in humans may be observed.

In LLMs, it is not only individual pieces of knowledge but the internal structure as a whole, formed through training, that may influence output. This point may be organized as a possible functional correspondence with the characteristic of L8, which treats the cognitive structure as a whole as a single structure.

However, this correspondence does not imply that humans and LLMs possess an identical cognitive structure. What this paper addresses is only the possibility of a functional correspondence in how output is produced.

This paper has examined the possibility of a functional correspondence between L8 and LLMs. The function in LLMs whereby the internal structure as a whole, formed through training, influences output is positioned as a candidate structural function corresponding to L8 in humans, and is referred to as the Structural Whole. This term is used to describe a functional correspondence; it does not refer to any specific implementation or technical component, nor does it imply that LLMs possess cognition or thought comparable to that of humans.

Conclusion

This paper has organized L8 as the cognitive layer that treats the cognitive structure as a whole as a single structure.

It was shown that L8 does not treat individual knowledge, experience, emotion, or theory as independent elements, but rather treats the cognitive structure as a whole — including all of these — as a single structure. It was also organized that this cognitive structure changes through interaction with other cognitive layers, and that its function is inferred through output that appears at Malkuth.

Furthermore, this paper examined the possibility of a functional correspondence with L8 not only in humans but also in LLMs, from the perspective that the internal structure as a whole, formed through training, influences output.

It should be noted that the function of L8 — treating the cognitive structure as a whole and determining that a state is ready for output — is related to Da'at, an integration point within SCOPE. However, how this integration comes to be realized is not addressed within the scope of this paper. The next paper will position Da'at as the integration point connecting the cognitive structure as a whole with output, and will examine its relationship with L8.